# Symmetry constraints on elliptic-curve scalar observables

Parity, normalized division polynomials, collision quotients, and Fourier null models

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**Status and reading guide.** This document records a human-initiated, substantially AI-developed mathematical investigation. The human curator directed the objectives but does not independently certify its advanced mathematical arguments. The novel claims and supplied proofs are candidate results awaiting independent specialist verification; public availability does not certify correctness, completeness or originality. This artifact develops scalar-parity symmetry restrictions, division-polynomial normalization identities, and a deliberately specified random-sign Fourier model. Elementary pairing and counting arguments are included, while normalization conventions, covariance calculations and the probabilistic limit should be checked independently. The surrogate model is not a measured property of deterministic elliptic-curve predicates, and no ECDLP hardness theorem or cryptanalytic speedup is proposed.

## Abstract

This manuscript investigates the constraints imposed by exact symmetries on public functions of elliptic-curve points when the corresponding scalar is unknown. In a cyclic group of odd prime order, a negation-invariant feature has one-half accuracy for canonical scalar parity under a symmetric distribution; a pairing argument is supplied. The treatment contrasts this obstruction with orientation-sensitive contact quantities and presents a proposed normalization identity for an intrinsic division-polynomial point function, distinct from a normalized elliptic divisibility sequence. The sixfold complex-multiplication quotient gives collision identities, but the idealized birthday gain is only a constant-factor effect. A coefficient-counting argument limits bounded-digit cyclotomic scalar representations. The manuscript also derives Fourier symmetry identities and studies a deliberately defined random-sign surrogate: its covariance is calculated and Gaussian marginal behavior is argued using characteristic functions. Parseval normalization, signed-orbit selection, and the limitations of Fourier-envelope deductions are treated separately. The candidate conclusions concern mathematical identities and the stated surrogate model, not benchmarks, deterministic parity leakage, or a general ECDLP lower bound.

**Keywords:** elliptic discrete logarithm; scalar parity; division polynomial; complex multipli-

cation; collision search; Fourier transform; symmetry.

## 1. Questions and mathematical setting

Let $E/\mathbf{F}_q$ be a nonsingular short Weierstrass curve in characteristic greater than three. Let $G$ generate a subgroup of odd prime order $n > 3$, and write $P_d = [d]G$. A scalar means its residue class modulo $n$ unless a canonical representative $0 \le d < n$ is explicitly specified. Coordinate arithmetic takes place in $\mathbf{F}_q$, not in the scalar field $\mathbf{F}_n$.

The exploratory investigation used numerical evaluations to test and refine the initial intuition about coordinate observables. This note isolates the exact identities and symmetry constraints underlying those questions. It makes no quantitative claim about a historical classifier, timing measurement, or numerical sample. The distinction is important: an algebraic obstruction, a deliberately specified random model, and a measured property of an actual deterministic predicate are different statements.

The standard theory of elliptic arithmetic, division polynomials, and collision algorithms is described in [1]. The relationship between scalar parity, normalized elliptic divisibility sequences, and ECDLP is prior work [2]. The role of efficient endomorphisms in known-scalar multiplication is likewise established [3]. We do not claim those general mechanisms as new.

## 2. Scalar division and an exact parity oracle

**Proposition 2.1 (scalar layers).** Let $m \ge 1$ be coprime to $n$. For $0 \le k < n$, the representative $x$ with $0 \le x < n$ of $km^{-1} \pmod n$ satisfies

$$x = \frac{k + \ell n}{m}, \qquad 0 \le \ell < m, \qquad k + \ell n \equiv 0 \pmod m. \tag{2.1}$$

**Proof.** The integer $mx - k$ is a multiple of $n$. It lies strictly above $-n$ and strictly below $mn$, so it is $\ell n$ with $0 \le \ell < m$. The congruence determines $\ell$ uniquely since $n$ is invertible modulo $m$. Conversely, that choice makes the quotient integral and between zero and $n-1$. □

In particular, for $m = 2$, even $k$ gives $x = k/2 < n/2$, whereas odd $k$ gives $x = (k+n)/2 > n/2$. This is a statement about the hidden integer representative. The easily computed point $[m^{-1}]P_k$ does not itself identify which scalar interval contains $x$.

**Proposition 2.2 (an exact oracle suffices).** An oracle returning canonical scalar parity for every point of $\langle G \rangle$ determines the complete scalar using at most $\lceil \log_2 n \rceil$ oracle calls and known group operations.

**Proof.** Starting from $Q = [d]G$, let $b \in \{0, 1\}$ be the returned parity and set $Q' = [2^{-1}](Q - [b]G)$. Since $d - b$ is even as an integer and $0 \le (d-b)/2 < n$, the canonical scalar of $Q'$ is exactly $(d-b)/2$, with no modular wrap. Repeat until the point is $O$. The recorded bits, in least-significant-first order, recover $d$. This is the standard parity-oracle reduction, also given in [2, Proposition 2]. □

A bit of an ordinate is not that oracle. Its usefulness requires a separate relation to the canonical scalar label.

## 3. Negation-invariant features cannot predict parity

**Proposed Theorem 3.1.** Let $L(P) = L(-P)$ on nonidentity subgroup points. Under the uniform distribution on nonzero scalars, every deterministic predictor depending only on $L(P_d)$ has parity accuracy exactly $1/2$. The same holds in expectation for a randomized predictor whose auxiliary randomness is independent of the hidden scalar. More generally, the conclusion holds for any distribution assigning equal mass to $d$ and $n - d$.

**Proof.** The involution $d \mapsto n - d$ pairs all nonzero scalars without fixed points. Because $n$ is odd, the canonical representatives in a pair have opposite parity. Their features are identical, so a deterministic prediction is correct on exactly one member. For a randomized predictor, condition on its random choices or pair its two equal prediction probabilities. Averaging proves the statement. □

The theorem applies to any vector of functions of $x(P)$ or $y(P)^2$, not only to one scalar feature. Undefined values must be excluded symmetrically. It does not apply to a nonsymmetric interval prior or to features containing orientation information. In particular, adding a known offset to a point can destroy the symmetry of a feature, but that fact alone does not imply predictive advantage.

## 4. Contact norms and an orientation-sensitive quantity

Write $E : y^2 = C(x)$ and let $A, B$ be fixed rational functions of $x$. For $h = A - By$ and $\bar{h} = A + By$,

$$N = h\bar{h} = A^2 - CB^2$$

is invariant under negation. Wherever the denominators are nonzero, define

$$J_h = \frac{d}{dx} \log \frac{h}{\bar{h}} = \frac{2C(A'B - AB') - ABC'}{y(A^2 - CB^2)}. \qquad (4.1)$$

**Derivation.** Since $y' = C'/(2y)$, expand $h'\bar{h} - h\bar{h}'$. The cross terms leave $2y(A'B - AB') - ABC'/y$. Dividing by $h\bar{h}$ yields (4.1). The numerator depends only on $x$, while the denominator contains one factor of $y$, so $J_h(-P) = -J_h(P)$.

The fixed-function qualification matters: if $A, B$ are themselves selected by a point-dependent rule, the whole rule's transformation under negation must be checked. For fixed $A, B$, the norm is subject to Theorem 3.1 and the oriented quantity is not. Neither statement supplies a correlation theorem for $J_h$.

## 5. An intrinsic normalized division-polynomial function

Assume in this section that $\gcd(n, q - 1) = 1$. Let $\psi_m$ be the standard division polynomials, with $\psi_1 = 1$, $\psi_2 = 2y$, and composition identity

$$\psi_{ab}(P) = \psi_a([b]P)\, \psi_b(P)^{a^2}. \qquad (5.1)$$

For nonzero $P \in \langle G \rangle$, define $L = q - 1$, choose $u$ with $un^2 \equiv 1 \pmod{L}$, and set

$$\Theta(P) = \left( \frac{\psi_L(P)}{\psi_{L+n}(P)} \right)^u. \qquad (5.2)$$

Both denominator and numerator are nonzero: their indices are not divisible by the prime order $n$. Exponentiation by $u$ is the unique inverse to the $n^2$-power map of $\mathbf{F}_q^*$. Thus (5.2) is a function of the point, with known indices independent of its unknown scalar.

**Proposed Theorem 5.1 (normalization identity).** For every integer $e$ not divisible by $n$,

$$\Theta([e]P) = \Theta(P)^{e^2} \psi_e(P). \qquad (5.3)$$

**Proof.** First take positive $e$; negative indices follow from $\psi_{-e} = -\psi_e$ and the same composition identity. Put $U = \psi_e(P)$. Since $[L+n]P = [L]P$, applying (5.1) in both orders gives

$$\frac{\psi_L([e]P)}{\psi_{L+n}([e]P)} = \left( \frac{\psi_L(P)}{\psi_{L+n}(P)} \right)^{e^2} U^{(L+n)^2 - L^2}.$$

The last exponent is congruent to $n^2$ modulo $L$. Taking the unique $n^2$th root proves (5.3). All cancellations are legitimate: $n$ divides neither $e$ nor $L$, so $\psi_e(P)$ and $\psi_e([L]P)$ are nonzero. □

To distinguish the conventions explicitly, the normalized sequence attached to a *fixed* base point is

$$\widetilde{W}_P(e) = \frac{\Theta([e]P)}{\Theta(P)} = \Theta(P)^{e^2-1} \psi_e(P), \qquad \widetilde{W}_P(1) = 1. \qquad (5.4)$$

The exponent in the intrinsic point-function identity is $e^2$; the exponent in the normalized sequence is $e^2 - 1$. The distinction is forced already by substituting $e = 1$. With $\Theta(O) = 0$, (5.4) is periodic in $e$ modulo $n$ and has zero terms exactly at multiples of $n$. The specific factor $\Theta(P)^{e^2-1}$ preserves the normalized Ward recurrence: both sides acquire the common factor $\Theta(P)^{2(m^2+n_0^2)-2}$ when its recurrence indices are $m, n_0$. This is the normalization mechanism studied in [2, Section 3]; the derivation above fixes the point-function and sequence conventions without relying on a formula inferred from numerical data.

Let $\chi_q$ be the quadratic character. If $\chi_q(\Theta(P)) = -1$, (5.3) gives

$$\chi_q(\Theta([e]P)) = (-1)^e \chi_q(\psi_e(P)). \qquad (5.5)$$

The left side is a public point function, but the unknown-index term on the right is not supplied by it. Knowing that term would provide scalar parity; merely computing $\Theta(Q)$ does

not. The character condition on $\Theta(P)$ is explicit, rather than assumed for every generator. The normalization and residue problems in [2] explain why this is a substantive missing input.

## 6. Sixfold symmetry and exact collision equations

Assume $E : y^2 = x^3 + b$ with $b \neq 0$, $q \equiv 1 \pmod 3$, and that the subgroup is preserved by $\phi(x, y) = (\beta x, y)$, where $\beta$ has order three. Write $\phi(G) = [\lambda]G$ and assume $\lambda$ has order three modulo $n$. These hypotheses hold for the standard secp256k1 group [4]. Put $H = \{\pm 1, \pm\lambda, \pm\lambda^2\}$. Every nonzero scalar orbit has six elements, since multiplication by a nontrivial unit of $H$ fixes no element of $\mathbf{F}_n^*$.

A collision state with public coefficients is $R_i = [A_i]G + [B_i]Q$, where $Q = [d]G$. If $R_i = [v]R_j$ for a known $v \in H$, then

$$(B_i - vB_j)d = vA_j - A_i \pmod n.$$

When $B_i - vB_j \neq 0$, this determines

$$d = (vA_j - A_i)(B_i - vB_j)^{-1} \pmod n. \qquad (6.1)$$

This follows by substituting $Q = [d]G$ and comparing coefficients of a generator. If the denominator is zero, a consistent collision also has zero numerator and provides no equation for $d$. Every proposed answer should be checked by $[d]G = Q$.

Canonicalizing a state by $R \mapsto [v]R$ requires changing its coefficients to $(vA, vB)$ at the same time. Otherwise the stored relation between point and coefficients is false.

### 6.1. The idealized birthday gain

For independent uniform samples from a set of $M$ states, the probability of no collision in the first $s$ samples is

$$\prod_{j=0}^{s-1}(1 - j/M).$$

For $s = O(\sqrt{M})$, expanding its logarithm gives $-s(s-1)/(2M) + O(s^3/M^2)$. Thus the collision scale is $\sqrt{M}$. Replacing $n - 1$ states by $(n-1)/6$ in this *sampling model* yields the factor $\sqrt{6}$ in sample count. It does not establish a measured running-time gain for a deterministic or Pollard-style quotient walk. Walk mixing, fruitless cycles, restarts, field costs, and coefficient bookkeeping require their own analysis [1].

Efficient-endomorphism decompositions $k \equiv a + \lambda b \pmod n$ can accelerate a known scalar multiplication $[k]P = [a]P + [b]\phi(P)$ [3]. Applying the same notation to an unknown $d$ does not reveal $a, b$ or produce two independent smaller discrete logarithms.

## 7. Bounded-digit representations need a coverage proof

**Proposition 7.1.** Let a field of odd characteristic contain an element $\gamma$ of even order $D$, and let $A_j \in \{0, \dots, L-1\}$. The number of distinct values represented by $\sum_{j=0}^{D-1} A_j \gamma^j$ is at most

$$(2L-1)^{D/2}. \qquad (7.1)$$

**Proof.** Since $\gamma^{D/2} = -1$,

$$\sum_{j=0}^{D-1} A_j \gamma^j = \sum_{j=0}^{D/2-1} (A_j - A_{j+D/2})\gamma^j.$$

There are at most $2L-1$ possible integer differences at each position. Reducing them into the field or identifying different vectors can only lower the number of outputs. □

For $D = 64$ and $L = 16$, the upper bound is $31^{32}$, fewer than $2^{159}$, rather than $16^{64} = 2^{256}$. This rules out full coverage of a roughly 256-bit scalar field by that particular digit box. Cycling coefficients can still give a valid multiplication identity; coverage does not follow from that identity.

## 8. Ordinate predicates and Fourier symmetry

Now let $q = p$ be an odd prime, let $E(\mathbf{F}_p) = \langle G \rangle$ have prime order $n$, and retain the order-three automorphism of Section 6. For canonical ordinate representatives $0 \le y < p$, three possible predicates are

$$f_{\text{bit}}(P) = (-1)^y, \qquad f_{\text{half}}(P) = \begin{cases} 1, & 2y < p, \\ -1, & 2y > p, \end{cases}$$

$$f_{\text{quad}}(P) = \chi_p(y(P)), \qquad f(O) = 0.$$

There is no rational nonzero two-torsion, so $y \neq 0$ away from $O$. All three predicates are invariant under $\phi$. The first two are odd under negation because the representative of $-y$ is $p - y$; the third is odd when $p \equiv 3 \pmod 4$. None of these identities is a scalar-parity prediction.

For any real sequence satisfying $f_0 = 0$, $f_{\lambda d} = f_d$, and $f_{-d} = -f_d$, define

$$S_k = \sum_{d=0}^{n-1} f_d e^{-2\pi i k d/n}.$$

Changing variables gives

$$S_0 = 0, \quad S_{\lambda k} = S_k, \quad S_{-k} = -S_k = \overline{S_k}. \qquad (8.1)$$

The spectrum is purely imaginary, and there are $(n-1)/6$ nonzero frequency orbits of equal magnitude. Equality of frequency values within an orbit is deterministic dependence, not evidence for independent observations.

## 9. A symmetry-preserving random-sign model

Choose one representative $d_j$ of each sixfold scalar orbit and independent signs $\varepsilon_j \in \{-1, 1\}$ of equal probability. Define $f_{\lambda^a d_j} = \varepsilon_j$, $f_{-\lambda^a d_j} = -\varepsilon_j$ for $a = 0, 1, 2$, and $f_0 = 0$. This is a specified surrogate, not an assertion that an arithmetic predicate is random.

**Proposition 9.1 (covariance).** For nonzero frequencies $k, l$,

$$\mathbb{E}S_k = 0, \qquad \mathbb{E}(S_k\overline{S_l}) = \begin{cases} n, & l \in \{k, \lambda k, \lambda^2 k\}, \\ -n, & l \in \{-k, -\lambda k, -\lambda^2 k\}, \\ 0, & \text{otherwise.} \end{cases} \tag{9.1}$$

**Proof.** For nonzero scalar indices, the input covariance is one when $e = \lambda^a d$, minus one when $e = -\lambda^a d$, and zero otherwise. Substitute this into the double Fourier sum. For each $a$, the sums over $d \neq 0$ are $n-1$ when the relevant frequency is zero and $-1$ otherwise. The two constant terms cancel, leaving

$$n \sum_{a=0}^{2} (\mathbf{1}_{l\lambda^a = k} - \mathbf{1}_{l\lambda^a = -k}).$$

The three multipliers are distinct and none is the negative of another, which proves (9.1). □

**Proposition 9.2 (Gaussian marginals of the surrogate).** Along any sequence of such prime orders $n \to \infty$, the real variable $S_k/(i\sqrt{n})$ converges in distribution to a standard normal variable for any choice of a nonzero frequency at each size.

**Proof.** The contribution of each scalar orbit is an imaginary number times its independent sign. Thus $S_k/i = \sum_j a_j \varepsilon_j$ with real $a_j$ and $|a_j| \leq 6$. Proposition 9.1 gives $\sum_j a_j^2 = n$. For fixed real $t$, the characteristic function is

$$\prod_j \cos(t a_j/\sqrt{n}).$$

Using $\log\cos u = -u^2/2 + O(u^4)$ uniformly near zero, its logarithm is

$$-\frac{t^2}{2} + O\left(\frac{t^4 \sum_j a_j^4}{n^2}\right) = -\frac{t^2}{2} + O(t^4/n),$$

since $\sum_j a_j^4 \leq 36n$. Convergence of characteristic functions proves the assertion. □

This is a theorem about the explicitly generated signs. It is not a Gaussian-limit theorem for ordinate parity, threshold predicates, or the quadratic ordinate character. Nor does

zero covariance between different frequency orbits make the finite-size Fourier coefficients independent.

## 10. Parseval, sign selection, and envelope consequences

For any deterministic sign sequence with $f_0 = 0$ and $|f_d| = 1$ for $d \neq 0$, Fourier orthogonality gives

$$\sum_{k=0}^{n-1} |S_k|^2 = n(n-1). \qquad (10.1)$$

When $S_0 = 0$, its nonzero-frequency average is exactly $n$. If $S_k$ is imaginary and $X_k = S_k/(2i\sqrt{p})$, then

$$\frac{1}{n}\sum_k X_k^2 = \frac{n-1}{4p}, \qquad \frac{1}{n-1}\sum_{k\neq 0} X_k^2 = \frac{n}{4p}. \qquad (10.2)$$

Thus a variance match is not an independent distribution test. Selecting one member from each sixfold frequency orbit preserves all even moments, but not the signed empirical law: choosing the positive imaginary member of each pair makes the selected values nonnegative. A canonical signed comparison uses the full spectrum or includes both signs for each orbit.

**Proposition 10.1 (consequences of an assumed envelope).** Suppose $|S_k| \leq B$ at every nonzero frequency and $S_0 = 0$. A Fourier-supported approximation capturing a fraction $\eta$ of the total energy must use at least $\eta n(n-1)/B^2$ frequencies. Also, for every fixed $u \neq 0, v \pmod n$,

$$\left|\frac{1}{n-1}\sum_{d=1}^{n-1}(-1)^d f_{ud+v}\right| \leq \frac{B(2+\log n)+1}{n-1}. \qquad (10.3)$$

**Proof.** A set of $s$ frequencies contains at most $sB^2$ energy; compare with (10.1). For the second assertion let $a_d = (-1)^d$ on $0 \leq d < n$. A finite geometric sum gives

$$|\widehat{a}(k)| = \frac{1}{|\cos(\pi k/n)|}.$$

Pair frequencies around $n/2$ and use $\sin x \geq 2x/\pi$ for $0 \leq x \leq \pi/2$. The resulting sum over half-integers gives

$$\sum_{k=1}^{n-1} |\widehat{a}(k)| \leq n(2+\log n).$$

The Fourier magnitudes of $d \mapsto f_{ud+v}$ are a permutation of those of $f$. Fourier inversion bounds the full scalar correlation by $B(2+\log n)$. Removing its $d = 0$ term changes it by at most one. Division by $n-1$ proves (10.3). $\square$

The envelope is an explicit hypothesis here, not an assumed property of every predicate. These conclusions concern sparse Fourier approximation and direct correlation with one fixed affine query. They do not rule out arbitrary adaptive combinations, other features, or non-Fourier algorithms.

## 11. Interpretation

The useful distinction is between information preserved by a symmetry and a method for extracting a hidden scalar. Negation-invariant features cannot predict canonical parity under symmetric sampling. Orientation-sensitive quantities escape that exact obstruction but require a further argument to become useful. Normalized division-polynomial point functions satisfy a precise identity, yet the unknown-index residue remains a separate input. Sixfold quotient collisions yield exact equations when coefficients are tracked, while their birthday benefit remains a constant-factor statement in an explicit model.

The Fourier null model demonstrates how Gaussian-looking marginal behavior can follow from symmetry-compatible random signs. It does not prescribe the law of a deterministic arithmetic observable. Numerical evaluations can motivate a sharper arithmetic theorem, but they neither replace that theorem nor establish a general security conclusion. The scope of every restriction in this note is therefore specified by its hypotheses rather than by a numerical observation.

## Research provenance, contributions, and verification status

### Origin and development of the investigation

This mathematical investigation was initiated and directed by Louis Desbiens and developed through extensive interaction with OpenAI's ChatGPT. Louis Desbiens identified the original research questions, set the objectives, guided the investigation, assessed its progress at a conceptual level, and selected the material for public dissemination.

ChatGPT contributed substantially to mathematical exploration, including alternative formulations, derivation of algebraic identities, formulation and examination of conjectures and proposed proofs, identification of potential counterexamples and exceptional cases, design and execution of symbolic and numerical checks, organization of relevant literature, and drafting and revision of the text. These contributions went well beyond linguistic or editorial assistance.

### Attribution and scientific authorship

Louis Desbiens is identified as the **human initiator and curator**, not as an independent mathematical authority certifying the results. OpenAI's ChatGPT is identified as the **AI mathematical research contributor**, not as an accountable conventional scholarly coauthor. These designations describe the provenance without implying that the human curator personally derived or independently verified all the arguments.

### Mathematical verification status

The candidate mathematical contributions have not undergone independent specialist validation, formal proof verification, or an exhaustive originality and priority review. In particular, independent examination is needed of the scope of the parity pairing; division-polynomial normalization and recurrence conventions; the random-sign covariance and Gaussian limit;

separating the assumed Fourier envelope from deterministic-curve conclusions. Locally displayed proofs are proposed arguments. Established results cited from public literature should be assessed separately from new deductions; neither an AI-generated proof nor a successful computational check is by itself an independently certified theorem. Individual statements may be correct, partly correct, incomplete, or incorrect.

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No institutional affiliation or endorsement is claimed. The investigation was undertaken independently, without institutional funding or resources. No external research funding was received. The human curator reports no competing interests.

### Computational evidence and source availability

Numerical evaluations helped develop the initial intuition. The public repository contains the available scripts, archived results, and a claim-by-claim verification coverage guide. These are scoped finite arithmetic, symbolic, and numerical checks, not independent verification of the general mathematical results. Historical exploratory calculations for which independent source data are unavailable are not treated as reproducible evidence.

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